

INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

INTRO TO CRITICAL INFRASTRUCTURE FOR IT LEARNERS

LAB 01 — INTRODUCTION TO PROGRAMMABLE LOGIC CONTROLLERS

Student Edition

Authorized ICDFA training environments only
icdfa.edu.ng

Lab 01: Introduction to Programmable Logic Controllers

Course: Intro to Critical Infrastructure for IT Learners | **Estimated Duration:** 2–3 hours

Submission deadline: As published in ICDFA E-Campus.

Authorization & Safety: Perform all activities only on the ICDFA-provided virtual machines, packet captures and isolated training networks. Do not bridge these VMs to production networks, scan external systems, or reuse credentials observed in training data.

Learning Outcomes

- Explain the role of PLCs in industrial and critical-infrastructure environments.
- Identify core PLC components including processor, power supply, input modules, output modules, rack/chassis and expansion modules.
- Differentiate discrete (digital) and analog I/O.
- Launch OpenPLC Editor and inspect a ladder-logic motor-control example.
- Explain the purpose of normally open contacts, normally closed contacts and output coils.

Laboratory Environment

System	Expected Address	Account	Password
PLC VM	192.168.1.10	plc	password

Note: If the instructor-provided VM image uses updated addressing or credentials, use the values published in E-Campus. Do not alter addressing unless instructed.

Pre-Lab Checklist

- Read the complete lab before beginning.
- Confirm you can access the required VM(s) and that virtualization support is enabled.
- Use the instructor-specified isolated/host-only network.
- Create or restore the clean starting snapshot for this lab.
- Prepare a report document with your name, registration number, course, lab number and date.
- Capture evidence as you progress rather than repeating the lab at the end.

Procedure

Step 1. Import the ICDFA-provided PLC VM into the virtualization platform specified by your instructor. Configure the laboratory network as Internal/Host-Only unless your instructor provides a different isolated configuration. Do not use Bridged networking.

Step 2. Create a clean snapshot named LAB01-START before making changes. Start the PLC VM and sign in with the assigned laboratory account.

Step 3. Confirm that the VM is using the expected training address 192.168.1.10. If your instructor supplied a revised image or address, record the value in your report instead of changing it without approval.

Step 4. Open the OpenPLC Editor from the desktop/application menu.

Step 5. Select File > Open and browse to the provided PLC Programming Language Examples directory. Open the Ladder Diagram project.

Step 6. Under the Project tree, open Res0 and edit instance0 (Motor_Control). Expand the ladder diagram so the full motor-control logic is visible.

Step 7. Identify the two power rails, the rung(s), the Start contact, Stop contact, MotorOutput contact used for latching, and the MotorOutput coil.

Step 8. Trace the logic from the left rail to the right rail. Explain what must be true for the motor output to energize and what causes it to de-energize.

Step 9. Record one example each of a discrete input, analog input, discrete output and analog output that could appear in an industrial process.

Step 10. Capture the required evidence, save your report, then restore or power off the VM as directed by the instructor.

Required Evidence & Submission

- Screenshot showing the OpenPLC Motor_Control ladder diagram.
- A labelled explanation of the Start, Stop, latch and MotorOutput elements.
- A completed I/O classification table with at least four industrial examples.
- A 150–250 word reflection on why network-connected PLCs create both operational benefits and cybersecurity considerations.

Recommended filename: <RegistrationNo>_LAB01_Introduction_to_Programmable_Logic_Controllers.pdf

1. Labelled Explanation of Motor Control Elements

- **Start Contact (Normally Open - NO):** When pressed, it closes the circuit and allows current flow, starting the motor.
- **Stop Contact (Normally Closed - NC):** Remains closed during normal operation. Pressing it opens the circuit and stops the motor.
- **Latch (Seal-in Contact):** An auxiliary MotorOutput contact connected in parallel with the Start contact. Once the motor starts, this contact closes and keeps the circuit energized after the Start button is released.
- **MotorOutput Coil:** The output coil that energizes the motor starter, causing the motor to run when all logic conditions are satisfied.

I/O Classification Table

I/O Type	Industrial Example
Discrete Input	Emergency Stop Push Button
Discrete Input	Limit Switch
Analog Input	Temperature Sensor (4-20 mA)
Analog Input	Pressure Transmitter
Discrete Output	Motor Starter Relay
Discrete Output	Indicator Lamp
Analog Output	Variable Frequency Drive (Speed Command)
Analog Output	Control Valve Position Signal

Reflection:

Network-connected PLCs provide significant operational benefits in modern industrial environments. They enable remote monitoring, centralized control, real-time data collection, predictive maintenance, and improved process efficiency. Operators can quickly identify equipment issues, monitor production performance, and make adjustments without being physically present at the control system. This connectivity helps organizations reduce downtime, improve reliability, and support better decision-making through industrial automation and data analytics.

However, network connectivity also introduces cybersecurity risks. Traditional PLCs were designed primarily for reliability and operational control rather than security. When connected to plant networks or enterprise systems, PLCs may become targets for cyberattacks, malware infections, unauthorized access, and network-based exploitation. A successful attack could disrupt industrial processes, damage equipment, affect product quality, or cause service outages in critical infrastructure sectors such as energy, water treatment, transportation, and manufacturing.

To reduce these risks, organizations should implement security controls such as network segmentation, strong authentication, regular patch management, access control policies, continuous monitoring, and secure remote-access solutions. Balancing operational efficiency with cybersecurity is essential because the availability, integrity, and safety of industrial control systems directly impact both business operations and public services.

Knowledge Check

1. What is the difference between an input module and an output module?

Response: An input module receives signals from field devices such as sensors, switches, and push buttons and sends them to the PLC processor. An output module receives commands from the PLC processor and controls field devices such as motors, relays, valves, and indicator lights.

2. Why is a normally closed contact commonly associated with stop or emergency-stop logic?

Response: A normally closed contact provides fail-safe operation. If power is lost, a wire breaks, or the emergency-stop button is pressed, the circuit opens and the equipment safely stops.

3. Give one example of a discrete I/O signal and one example of an analog I/O signal.

Response: Discrete I/O: Push button ON/OFF signal. Analog I/O: Temperature sensor transmitting a continuous value such as 4-20 mA.

4. What operational risk can arise when industrial automation devices become network-connected?

Response: Network-connected industrial devices may be exposed to cyberattacks or unauthorized access, which can disrupt operations, cause equipment damage, or lead to loss of system availability.